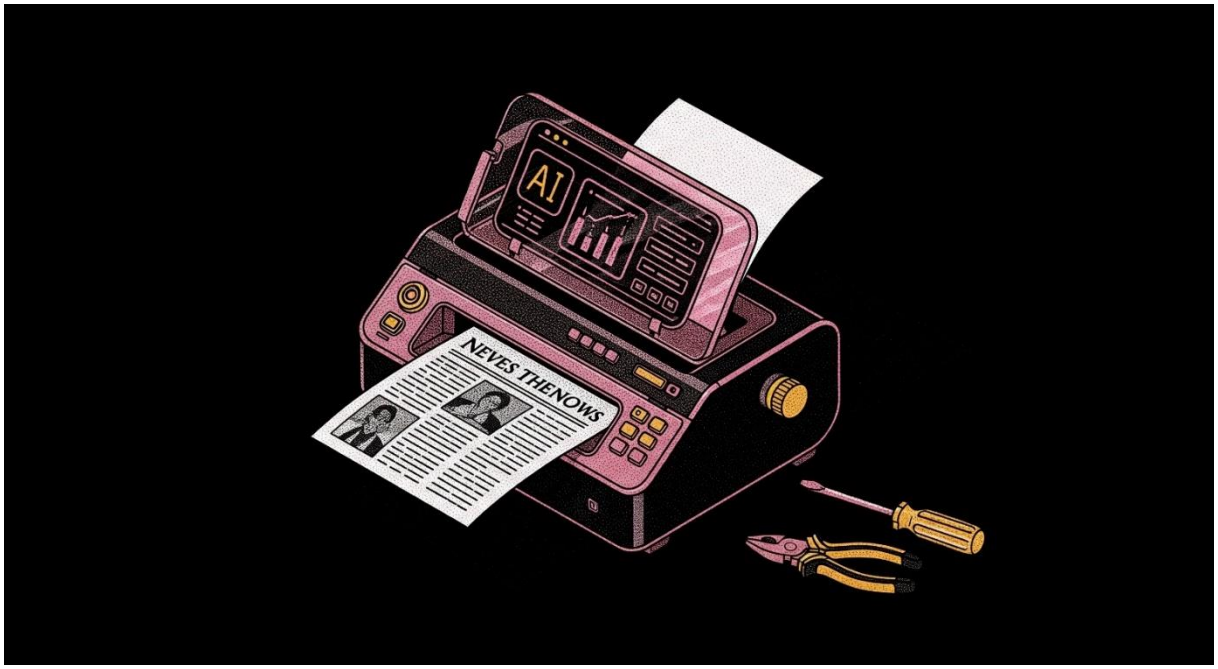


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Tailoring Transparency: Adaptive UX Strategies for Generative AI in Journalism

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Tailoring Transparency: Adaptive UX Strategies for Generative AI in Journalism

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This thesis examines how adaptive user experience strategies can improve transparency around generative AI in journalism, with the dual goals of strengthening reader trust and preserving editorial integrity. Conducted in collaboration with Schibsted ASA, the study followed a three-phase, mixed-methods design that included speculative design exercises to explore future transparency scenarios. Current transparency practices in AI-driven journalism are largely declarative and static, relying on simple “AI-generated” labels that do not clarify how or why AI contributed to the final output. This limitation weakens accountability and reduces user trust, which motivates the research presented in this thesis. Semi-structured interviews with newsroom editors, UX designers, and AI specialists identified core challenges and user requirements, which informed the creation of three interactive prototypes: a CMS disclosure builder, an AI “show thinking” chatbot, and a customizable content settings panel. Each prototype was evaluated through expert heuristic review, think-aloud testing with editors, and a value-alignment assessment to examine usability, workflow fit, and trust effects. Findings show that progressive disclosure controls and provenance indicators help balance cognitive load with editorial oversight, while chain-of-thought explanations enhance verifiability but require personalization safeguards. In contrast, a generic settings panel without provenance cues reduced both trust and perceived control. Contributions include empirical evidence of transparency–usability trade-offs, a replicable multimethod evaluation protocol, and three design strategies for newsroom integration. The thesis concludes with practical guidelines—on-demand “how it was made” expanders that clarify AI contributions to both text and images, and plain-language disclaimers that emphasize continued human editorial oversight and outlines future work on large-scale reader testing, live CMS deployment, operational impact metrics, and context-sensitive disclosure triggers.

SAMMANFATTNIN

Denna avhandling undersöker hur adaptiva strategier för användarupplevelse kan förbättra transparensen kring generativ AI inom journalistik, med de dubbla målen att stärka läsarnas förtroende och bevara redaktionell integritet. Studien, som genomfördes i samarbete med Schibsted ASA, följde en trefasdesign med blandade metoder som inkluderade spekulativa designövningar för att utforska framtida transparenspraxis. Nuvarande transparenspraxis inom AI-driven journalistik är till stor del deklarativ och statisk och förlitar sig på enkla "AI-genererade" etiketter som inte förtydligar hur eller varför AI bidrog till slutresultatet. Denna begränsning försvagar ansvarsskyldigheten och minskar användarnas förtroende, vilket motiverar forskningen som presenteras i denna avhandling. Semistrukturerade intervjuer med redaktionsredaktörer, UX-designers och AI-specialister identifierade centrala utmaningar och användarkrav, vilket informerade skapandet av tre interaktiva prototyper: en CMS-upplysningsbyggare, en AI-"show thinking"-chatbot och en anpassningsbar panel för innehållsinställningar. Varje prototyp utvärderades genom expertheuristisk granskning, högtänkande tester med redaktörer och en värdejusteringsbedömning för att undersöka användbarhet, arbetsflödesanpassning och förtroendeeffekter. Resultaten visar att progressiva kontroller av öppenhet och proveniensindikatorer hjälper till att balansera kognitiv belastning med redaktionell tillsyn, medan tankekedjans förklaringar ökar verifierbarheten men kräver personaliseringsskydd. Däremot minskade en generisk inställningspanel utan provenienssignaler både förtroende och upplevd kontroll. Bidragen inkluderar empiriska bevis för avvägningar mellan transparens och användbarhet, ett replikerbart protokoll för utvärdering med flera metoder och tre designstrategier för integration i nyhetsrum. Avhandlingen avslutas med praktiska riktlinjer – expanderande funktioner för "hur det gjordes" på begäran som förtydligar AI-bidrag till både text och bilder, och friskrivningar i tydligt språk som betonar fortsatt mänsklig redaktionell tillsyn och beskriver framtida arbete med storskalig läsartestning, live CMS-distribution, operativa effektmått och kontextkänsliga utlösare för öppenhet.

CCS Concepts: • Human-centered computing → Human computer interaction (HCI) → HCI design and evaluation methods →User studies

Keywords: generative AI, AI transparency, adaptive transparency, AI disclosure

Nyckelord: generativ AI, AI-transparens, adaptiv transparens, AI-avslöjande

1 INTRODUCTION

In recent years, the rapid evolution of Artificial Intelligence (AI) has significantly transformed the field of journalism. Innovations in generative AI have enabled automated content creation, personalized news delivery, and dynamic data analysis, providing newsrooms with unprecedented opportunities for efficiency and scalability [1]. However, these technological advances also bring forth complex challenges, particularly in ensuring transparency, maintaining editorial oversight, and upholding ethical standards in news production [1]. As AI systems increasingly mediate the creation and dissemination of news, they can also produce factual errors or "hallucinations," obscure the basis of their outputs, and blur distinctions between human and machine contributions [1, 2]. Because journalism plays a central role in shaping public knowledge, these technical problems immediately raise ethical questions about accountability, editorial responsibility, and whether audiences can continue to trust content that is shaped or assisted by algorithmic systems [2]. Addressing these concerns requires not only technical solutions but also design approaches that deliberately surface how AI contributes to published content [3].

Scholarly work highlights the importance of transparency and interpretability in AI applications. One study [4] emphasizes that explainability is vital for building user trust and supporting responsible use in critical settings [2]. Another study [5] finds that post-hoc explanation techniques clarify AI decisions, with direct relevance for automated journalism [6]. Meanwhile, adaptive User Experience (UX) approaches tailor transparency to different audiences [3, 7]. Drawing on adaptive hypermedia [8] and human-centered AI design [9], adaptive UX presents complex AI processes in ways that are accessible and informative. These strategies help news organizations adjust transparency to user experience levels, reducing cognitive load and avoiding information overload [8]. The Paradox of Disclosure further notes that both too much and too little transparency can undermine trust [10]. Moreover, the ambiguity of regulatory frameworks such as the EU AI Act [11] highlights the urgency of developing newsroom-centered approaches to disclosure. Prior work identifies key models—adaptive hypermedia [8], human-centered design [9], disclosure models like 5W1H [1], post-hoc explanation techniques such as LIME [5], and the Paradox of Disclosure [10]—but shows limited evidence for how to use these principles practically in editorial tools and newsroom workflows. This thesis addresses this gap by speculatively designing and short-term testing adaptive transparency strategies in collaboration with a leading media organization, without conducting long-term deployment in live newsroom platforms.

This research, conducted in partnership with Schibsted ASA, aims to address a central question: How can adaptive UX strategies enhance AI transparency in journalism, thereby building user trust and upholding journalistic ethics? To address this question, the thesis proceeds in four stages: (1) foundational research through expert interviews and thematic analysis [12], (2) ideation and prototype development, (3) evaluation of three prototypes via heuristic review [13, 14], think-aloud sessions [4], and a short-term value-alignment checklist (conceptually grounded in the Paradox of Disclosure [10]), and (4) synthesis of findings into a five-pillar conceptual framework and practical newsroom guidelines. Semi-structured interviews with nine Schibsted newsroom experts (editors, UX designers, AI specialists) were coded using a combined deductive–inductive thematic analysis [12] (deductive: applying predefined literature-derived codes to structure the analysis; inductive: enabling new codes and themes to emerge from the data). Insights from this phase informed the creation of three high-fidelity prototypes: a CMS disclosure builder, an AI "show thinking" chatbot, and a customizable content settings panel. Each prototype was then evaluated through heuristic review [13, 14], think-aloud sessions with professional editors [5], and a bespoke value-alignment checklist [10]. In addition to empirical evaluations, speculative design [15] is incorporated to envision alternative futures and stimulate critical reflection on transparency and accountability in AI-driven journalism.

The findings indicate that Prototype 1's progressive disclosure controls and provenance cues achieved an effective balance between cognitive load and editorial oversight. Prototype 2's chain-of-thought explanations enhanced verifiability but highlighted the need for clearer personalization boundaries. Prototype 3's generic settings panel was less effective, as vague disclosures diminished both trust and perceived control. Across all prototypes, the Paradox of Disclosure was evident: increased transparency often resulted in usability trade-offs. The thesis contributes (1) empirical validation of transparency–usability trade-offs in journalistic UX, (2) a replicable multimethod evaluation protocol for assessing adaptive transparency prototypes, and (3) three actionable design strategies for newsrooms to implement on-demand 'how it was made' expanders and humanized AI disclaimers. These findings offer practical recommendations for integrating AI in ways that reinforce, rather than compromise, human editorial judgment and ensure AI remains a supportive tool in digital newsrooms.

Future research should include large-scale, demographically diverse reader studies, deployment of prototypes in live CMS environments, development of operational impact metrics for AI interventions, and investigation of context-sensitive disclosure triggers aligned with natural reading behaviors. Large-scale reader studies are necessary to assess generalizability across audience groups and demographic segments [3]. Prototype deployment in live CMS environments will help evaluate ecological validity and workflow integration [16]. Operational impact metrics will allow newsrooms to quantify the editorial and audience effects of transparency interventions [7]. Context-sensitive disclosure triggers are promising because they can present explanations when readers need them most, reducing cognitive load and improving access to deeper information [8].

2 BACKGROUND

The rapid integration of AI into journalism is transforming news production and distribution. Automated content generation, personalized news delivery, and algorithm-driven reporting enhance efficiency and engagement but also introduce challenges, especially regarding transparency, accountability, and ethical journalistic practices. Moreover, studies [3, 7] highlight that mere labelling of AI-generated content is insufficient, calling for more nuanced disclosure practices. Different types of conceptual resources inform this discussion. Frameworks such as AI transparency models, including the 5W1H disclosure structure [1], provide an overarching way of organizing disclosure practices. Design approaches, for example human-centered design [9] and adaptive hypermedia [8], guide how these practices can be aligned with user needs and media literacy levels. Specific techniques and methods, such as post-hoc explanation tools like LIME [5], operationalize transparency at the level of system outputs. Finally, theoretical perspectives such as the Paradox of Disclosure [10] highlight the risks of both over- and under-disclosure. Grounding this thesis in this combination of frameworks, approaches, and methods demonstrates the need for adaptive strategies that reconcile technological innovation with rigorous editorial standards. The sections that follow elaborate on these distinctions in more detail, showing how they inform the design of UX solutions that support both transparency and accountability in AI-driven journalism.

2.1 Defining Key Terms

This section establishes the foundation of the study by contextualizing the rapid integration of AI into journalism and by clearly defining key terms critical to understanding the research.

- **Newsrooms**: A newsroom is the central hub within a media organization where journalists, including reporters, editors, and producers, collaborate to gather, produce, and manage news content for various platforms such as newspapers, online publications, radio, and television. This space serves as the nucleus for editorial decision-making, content creation, and the coordination of news dissemination. Beyond its physical structure, the newsroom embodies a cultural environment that builds journalistic identity and a sense of professional community [16].
- **Generative AI:** Generative AI refers to algorithms that produce new content, such as text, images, or audio, by learning patterns from existing data. A seminal work in this field on the paper on Generative Adversarial Networks, which laid the foundation for many modern generative models [17].
- **AI Transparency** AI transparency involves the clear communication of the functioning, decision-making processes, and limitations of AI systems. This concept is vital for building user trust and ensuring accountability. Several frameworks were discussed for interpreting complex models that are essential in high-stakes applications such as automated journalism [4, 18].
- **Adaptive UX:** Adaptive User Experience (UX) refers to design strategies that dynamically adjust interfaces and information presentation based on user behaviours, preferences, or expertise. Adaptive hypermedia demonstrates how tailoring content can enhance relevance enhance clarity [8].
- **Interpretability in AI:** Interpretability is the degree to which a human can understand the internal mechanics of an AI system or how an AI system thinks and makes decisions, and provides frameworks that stress the importance of making AI decisions accessible, which is crucial for verifying and validating AI-generated outputs in journalism [5, 18].

2.2 Ethical and Editorial Challenges in Journalism

There are ethical dilemmas introduced by AI in journalism [1] one of the most prominent ones is the “black box” nature of AI systems which complicates the tracing decision-making process, which challenges the assignment of editorial responsibility [2]. This opacity does not only harm accountability but also increases the risk of AI hallucinations and misleading outputs that

can undermine public trust. As the content becomes more and more AI-generated by the day it is crucial to authenticate information, disclosure practices can contribute to the spread of misinformation [1, 4]. Simple "AI Labels" may fail to capture this complex interplay between human and machine-generated content, thus amplifying the audience's scepticism [3]. This calls for more nuanced disclosure strategies that communicate the role of AI while preserving robust editorial oversight. This delicate balance between clarity and comprehensiveness is critical, as evidenced by findings [7]. AI can inadvertently compromise journalistic roles by shifting key editorial decisions from humans to algorithms [2]. This "erosion of professional agency" challenges the independence of journalists and raises critical questions about editorial responsibility. Addressing this challenge is pivotal; developing strategies that reinforce human oversight and preserve the distinctive expertise of journalists ensures that AI serves as a supportive tool rather than replacing human judgment.

The intersection of AI, ethics, and journalism extends beyond newsroom practices and influences the broader media ecosystem [2]. While automation enhances efficiency, it also requires vigilant editorial oversight to prevent risks associated with unchecked algorithmic processes, such as factual errors and opaque decision-making [1, 2]. Ethical deployment is therefore central to sustaining public trust, since opaque systems can increase the likelihood of misinformation and weaken perceptions of news credibility [1]. At the regulatory level, the ambiguity of frameworks such as the EU AI Act [11] further underscores the importance of transparent, user-centric disclosure strategies. In addition, automation raises concerns about job displacement and its impact on editorial decision-making; reduced human involvement may inadvertently diminish the diversity of perspectives and reinforce filter bubbles, limiting the viewpoints available to audiences [6]. This amplification of confirmation biases further complicates the ethical landscape, highlighting the need for robust and enforceable frameworks for responsible AI integration in journalism [2].

2.3 AI Transparency and Disclosure

While transparency is essential for maintaining trust, the reviewed studies provide limited detail beyond noting that AI's opacity complicates disclosure practices. Readers must be aware when engaging with AI-generated content, yet inconsistent labelling across media organizations leads to confusion [1, 2]. At its core, the concept of AI transparency is not simply about labelling content as "AI-generated" but involves communicating the specific role and limitations of AI within the journalistic process [7]. Transparency and disclosure are critical as they directly influence audience trust and the perceived credibility of news content [3].

The distinction between AI-generated and human-created content is not always clear-cut. Arguments are made regarding the traditional journalistic autonomy that must evolve into "authorial autonomy", where journalists remain responsible for the final output, but their creative agency is shaped by AI integration [7]. This shift raises questions about when AI's involvement is significant enough to need disclosure. The challenge lies in determining which aspects of content production are substantially influenced by AI and how this should be communicated to audiences [2]. Challenges in labelling and disclosure arise from inconsistent transparency practices across media organizations, leading to audience confusion and varying perceptions of credibility [1, 2]. Research [3] indicates that news consumers overwhelmingly demand detailed disclosures about AI's role in news production, yet simplistic "AI-generated" labels can have unintended consequences. If too vague, they risk undermining credibility by failing to clarify AI's involvement in content creation, while overly technical explanations may alienate audiences unfamiliar with AI's implications. To address these challenges, effective transparency strategies should move beyond generic labels to explicitly communicate how AI was used, why it was applied, and the extent of human oversight in the process. A study proposes a 5W1H (Who, what, When, Where, why, How) framework to structure AI disclosures in journalism [3], offering a balance between clarity and editorial flexibility. Its comprehensive nature may be impractical in digital-first formats where brevity is key, potentially overwhelming audiences with excessive details [1]. Complex disclosures risk increasing cognitive load, making AI transparency less accessible to non-expert readers [3]. Additionally, not all AI applications require the same level of disclosure; rigid adherence could lead to unnecessary or redundant explanations, making it difficult for journalists to tailor transparency efforts based on context [2]. AI's involvement in content creation, with overly technical explanations, may alienate audiences unfamiliar with AI's implications. To address these challenges, effective transparency strategies should move beyond generic labels to explicitly communicate how AI was used, why it was applied, and the extent of human oversight in the process

2.3.1 Industry Perspectives and Best Practices for AI Transparency

AI disclosure should be guided by its "significant journalistic impact" rather than a blanket approach [3]. This means that news organizations should develop adaptable and iterative disclosure mechanisms, balancing transparency with usability.

Similarly, legal frameworks such as Article 52 of the EU AI Act aim to provide AI transparency obligations. Still, their language remains ambiguous as terms like "authentic" and "significant impact" lack clear definitions, leaving room for interpretation [11]. Beyond legal obligations, recent research suggests that transparency should prioritize explainability over labelling [4]. Studies on interpretable AI emphasize that users are more likely to trust AI-assisted content if they understand how decisions are made [5]. Local Interpretable Model-Agnostic Explanations (LIME) provide a potential solution, offering human-readable justifications for AI outputs by approximating models locally [5]. This principle could be applied to journalism by developing interactive transparency mechanisms that allow users to explore AI's role in content generation dynamically.

The one-size-fits-all transparency approach is ineffective, emphasizing the need for adaptive transparency that adjusts based on user needs and media literacy levels. [6] Progressive disclosure enables users to access AI explanations at varying depths, preventing information overload while catering to different expertise levels [9]. Providing user control and personalization, such as the ability to filter AI-generated content or explore AI's role in news production, can reinforce trust and engagement [7]. Additionally, personas can help tailor AI transparency to different audience segments, ensuring clarity without overwhelming users [2]. However, ensuring effective AI transparency is only part of the solution; users must be able to interpret and engage with the disclosed information in a meaningful way. Simply offering transparency does not guarantee trust or comprehension, making it essential to present AI explanations in an accessible manner [2]. This highlights the role of UX strategies in AI-driven journalism, bridging the gap between complex AI processes and audience understanding

2.3.2 Editorial Policies and Audience Expectations

Empirical research into the influence of AI on newsroom practices reveals significant trends in both editorial policies and audience expectations regarding AI transparency [1, 3]. Current findings indicate that newsroom approaches to AI disclosure vary widely. Some organizations enforce strict disclosure standards, mandating comprehensive labels for AI-generated content, while others opt for a minimalist approach, disclosing only when AI has a "significant journalistic impact." This divergence reflects ongoing internal debates about balancing editorial control with the need for openness.

At the same time, surveys consistently show a strong demand for transparency among news consumers. For instance, 93.8% of respondents expect clear AI disclosures, and 91.5% believe that human editors should review AI-generated content. Over half of those surveyed prefer detailed explanations regarding both the rationale behind and the methodology of AI usage in news production [3]. Collectively, these insights underscore that effective AI transparency in journalism is a balancing act. Newsrooms must develop adaptive, user-centered policies that tailor disclosure levels to the significance of AI's role in content production while aligning with audience expectations. By continuously refining these practices in response to technological advancements and feedback, media organizations can mitigate risks associated with AI integration, reinforce credibility, and maintain public trust.

2.4 Interpretable Machine Learning and Trust

Many AI systems lack explainability mechanisms, making it difficult for both journalists and audiences to understand how AI-generated outputs are created [2]. Interpretability is essential for ensuring that both journalists and audiences understand the rationale behind AI-driven decisions. Studies [4] emphasize that a clear explanation of AI outputs is a fundamental requirement for responsible deployment in high-stakes areas like journalism, where accountability and credibility are paramount. Studies [5] argue that journalists must be able to verify AI-generated content before publication to prevent misinformation and bias. Evidence exists that transparency-driven environments can improve the usability of AI systems, making explanations accessible and reinforcing editorial control [19]. Interpretable AI is essential not only for journalists but also for news consumers who engage with AI-generated content. Audience-facing interpretability features, such as confidence scores, can help users assess the reliability of AI-generated news [18]. Transparency mechanisms, like editorial revision histories that highlight human edits to AI-generated content, clarify the extent of AI involvement in news production [7]. Additionally, progressive disclosure models allow users to explore AI explanations gradually based on their interests or media literacy, preventing information overload while maintaining transparency [3]. By presenting AI processes clearly and interactively, user-centered design strategies can enhance engagement, strengthen trust, and contribute to a more accountable and transparent journalistic process.

To address these challenges, researchers have proposed several methods to improve AI interpretability in journalism. One approach is post-hoc explanations, where techniques like Local Interpretable Model-Agnostic Explanations (LIME) generate simplified surrogate models to help journalists inspect AI-generated recommendations and identify potential biases or

inaccuracies before publication [5]. Additionally, transparency-driven AI environments that embed real-time, interactive explanations within editorial tools help journalists evaluate AI-driven classifications in context [7]. Human-AI collaboration models further contribute by incorporating human-in-the-loop systems, ensuring that AI-generated content is continuously refined by journalists rather than replacing critical editorial judgment [2]. Despite these promising strategies, significant challenges remain. The widespread reliance on black-box models creates a trust deficit, as both journalists and audiences struggle to understand or challenge AI-generated outputs [1, 2]. Opaque AI models, especially when employed in tasks such as fact-checking or bias detection, can increase skepticism among news consumers. Hidden biases in these systems further risk reinforcing systemic issues like echo chambers and ideological polarization, while a lack of clear attribution complicates error identification and accountability [3, 7]. Although current transparency efforts often emphasize labelling AI-generated content, there is a pressing need for more interactive explanations, such as showcasing how human editors modify AI outputs to further enhance trust and accountability in AI-driven journalism.

2.5 User Experience (UX) in AI Systems

User experience (UX) is a pivotal factor in determining how audiences engage with AI-driven journalism. Effective UX design not only enhances satisfaction through personalization but also plays a crucial role in mitigating risks such as filter bubbles and information overload [8, 9]. This theme examines the role of adaptive UX strategies in facilitating transparent, human-centred AI journalism, drawing primarily on the research conducted [2, 18, 21]. Adaptive UX strategies address this challenge by dynamically adjusting the level of AI transparency based on users' media literacy and engagement levels. These approaches ensure that essential details are accessible without causing information to overload [21]. By aligning the presentation of AI processes with user expectations, adaptive UX contributes to a more intuitive and trustworthy interface [8]. The emerging paradigm of "User Experience 3.0" emphasizes human-centred design, granting users control over transparency settings so they can customize the depth of information they receive about AI operations. This prevents excessive disclosure that might otherwise reduce trust [7, 9]. Building on both historical and contemporary insights into intelligent user interfaces [20], a promising technique not inundated with technical information upfront, while still having the opportunity to explore AI processes in greater depth if desired.

The adaptive UX strategies proposed in the literature emphasize methodological enhancements to improve AI transparency in journalism. Empirical evaluations, such as usability testing and A/B experiments, will assess how different transparency settings influence user trust and engagement, measuring users' ability to navigate and customize these features effectively [1, 24]. Iterative prototyping, guided by qualitative user interviews and think-aloud sessions, ensures that the interface meets diverse user needs without overwhelming them with information [2, 5, 25]. Additionally, controlled experiments will evaluate the effectiveness of progressive disclosure mechanisms by comparing user comprehension and satisfaction with incremental transparency versus static, all-at-once disclosure formats [3, 14]. The selection of these methods reflects their dual suitability for practical newsroom application and rigorous evaluation. Usability testing and A/B experiments provide quantifiable evidence on how variations in transparency impact user trust and comprehension [1, 24], while qualitative approaches such as think-aloud protocols and interviews reveal nuanced user expectations and cognitive processes [5, 25]. Progressive disclosure and iterative prototyping were emphasized over other HCI methods because they directly address the central challenge in AI-driven journalism: maintaining credibility while minimizing cognitive overload [3, 14]. Collectively, these strategies illustrate possible directions for designing adaptive transparency. In this thesis, selected methods such as expert interviews, heuristic evaluation, and think-aloud protocols were applied, while others such as A/B testing and large-scale controlled experiments are discussed as future work.

2.6 Speculative Approaches for Adaptive AI Transparency in Journalism

Speculative design is a forward-thinking methodology that challenges conventional design practices by exploring alternative futures and questioning prevailing assumptions [22]. Unlike traditional design methods, which tend to address clearly defined, present-day problems, speculative design is geared toward provoking critical dialogue about the broader societal, ethical, and cultural implications of emerging technologies [1]. A key aspect of this approach is its ability to generate design fiction as hypothetical scenarios that help make abstract future possibilities more tangible. Crafting the speculation enables designers to articulate nuanced futures that reveal the potential impacts and challenges of technological innovations [23]. In this way, speculative design acts as a catalyst for critical reflection, drawing attention to issues like transparency and accountability within sectors experiencing rapid change, such as AI-driven journalism. Some argue that speculative design opens a conceptual space where established boundaries can be reconsidered; by inviting stakeholders to engage with alternative

scenarios, speculative design encourages a deeper examination of the ethical and practical dimensions of emerging technologies [15].

Incorporating speculative design within this study offers several notable advantages. Firstly, by envisioning alternative futures of AI integration in journalism, by applying speculative methods, this study explores latent challenges related to transparency, accountability, and editorial oversight [23]. Secondly, the generation of design fictions nurtures critical discourse among stakeholders, thereby encouraging an examination of conventional narratives and the exploration of innovative disclosure strategies [15]. Lastly, the insights derived from speculative narratives serve as a conceptual counterbalance to empirical user-centered methods, ensuring that while the study rigorously addresses practical concerns through empirical evaluation, it simultaneously remains sensitive to long-term ethical and societal implications [22].

2.7 Research Gaps

While the previous sections review existing literature on AI transparency, interpretability, and adaptive user experience, a clear research gap remains in understanding how transparency can be operationalised through user-centred design within newsroom tools. Existing studies acknowledge the importance of explainability and disclosure but rarely test these principles empirically in professional editorial contexts. Much of the prior research focuses on audience perception or ethical reasoning rather than practical integration in design. This gap highlights the need for multimethod, design-based inquiry that connects theoretical principles of trust and ethics with functional newsroom applications.

3 METHODOLOGY

To define the scope and direction of this study, the research question draws on key challenges observed in current newsroom practices, audience expectations, and evolving standards around AI explainability. These challenges highlight persistent tensions between the need for transparency and the risk of information overload, as well as the gap between ethical imperatives and practical implementation. In particular, the increasing reliance on generative AI in journalism raises questions about how disclosure strategies can be made both responsible and adaptive to users' needs. These considerations form the basis for the research inquiry outlined below.

The primary research question guiding this study:

How can adaptive UX strategies for generative AI in journalism effectively tailor transparency to user needs, thereby navigating the inherent "paradox of disclosure" [10] (disclosing and not disclosing diminish trust) to build and maintain trust?

Sub-questions include:

1. What are the challenges and opportunities for AI disclosure in newsrooms?
2. How do expert perspectives from UX, AI transparency, and journalism inform effective disclosure models?
3. What design elements best support a balanced, user-centered approach to transparency?

This study adopts a hybrid methodological approach that integrates both speculative design and functional, user-centered design [9] methods. In this study, speculative design is employed during the early stages to generate alternative narratives and scenarios for how AI transparency might evolve, allowing for a critical examination of potential challenges and opportunities. In contrast, traditional user-centered design focuses on grounding design decisions in real user needs and behaviors. Guided by principles of human-centered design as highlighted in recent UX literature [9] this approach informs the development of a functional prototype. Trust and ethics were operationalised through both design and evaluation procedures. Trust was examined as a construct emerging from perceived editorial control, verifiability, and cognitive manageability, consistent with the five pillars developed in the foundational phase. Ethical considerations were embedded through continuous attention to accountability and informed consent, ensuring that AI disclosures and editorial oversight were treated as inseparable elements of responsible design. In practice, ethical integrity was maintained by emphasizing human-in-the-loop workflows and by treating transparency not only as a functional requirement but also as a normative standard guiding prototype development and evaluation.

By incorporating empirical insights from expert interviews, the functional design phase ensures that the prototype effectively addresses practical UX challenges related to AI transparency. This approach allows the study to balance imaginative exploration with rigorous evaluation: speculative design enables the envisioning of future-oriented scenarios [22], while user-centered design ensures that solutions are practical, accessible, and directly relevant to the needs of journalists and audiences

[9]. The iterative nature of the methodology directly addresses the complexity and potential unpredictability inherent in designing for AI transparency. Early findings, such as the identification of the paradox of disclosure [10], could thus be systematically fed back into the design process, leading to the development of more relevant and effective prototypes. This feedback loop between research phases strengthens the study by ensuring its continuous responsiveness to the data being collected, ultimately positioning the research to produce not just theoretical understanding but also practical, evaluable design solutions [24]. The research was conducted in three phases (see Figure 1).

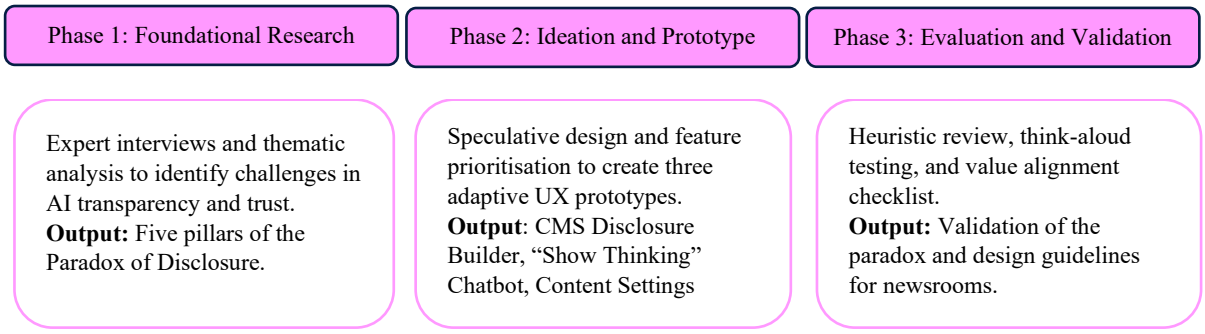


Figure 1: Overview of the research process showing Phase 1 (Foundational Research), Phase 2 (Ideation and Prototype Development), and Phase 3 (Evaluation).

3.1 Phase 1: Foundational Research

The initial phase of the research was dedicated to an in-depth exploration aimed at identifying and understanding the core challenges, tensions, and complexities associated with implementing AI transparency in contemporary journalistic practices [1, 2]. This foundational work was crucial for framing the problem space and ensuring that subsequent design and development efforts were grounded in real-world needs and expert insights [3, 7]. To gather rich qualitative data, semi-structured interviews were conducted with a cohort of domain experts (see Table 1). This method is well-established for exploring complex topics where individual perspectives, experiences, and tacit knowledge are paramount. The interviews aimed to uncover tensions and trade-offs concerning AI transparency within news media, including the practical feasibility of transparency measures, current industry practices, audience understanding and expectations, and the evolving legislative landscape [1]. Within this foundational phase, trust was interpreted as the user's confidence in both the process and the provenance of AI-assisted content. Ethics referred to the preservation of editorial accountability and human oversight. These interpretations informed both the interview protocol and the coding framework, ensuring that participants' reflections on trust and ethics were translated into tangible design criteria for adaptive transparency

Participants were recruited from Schibsted, a prominent media organization, to provide a holistic view of the challenges across the news production ecosystem [16]. They represented three key stakeholder groups: journalists and editors, who offered insights on editorial impacts, workflow changes, and maintaining journalistic standards [1]; AI transparency and AI experts, who provided technical perspectives on disclosure mechanisms, limitations of AI models, and explaining "black box" systems [18]; and UX designers and product managers, who contributed expertise on user-facing design, communicating complex information without cognitive overload [9]. Table 1 presents participant information (excluding names), including roles, affiliations, and experience.

Table 1: Participant information

Age	Gender	Country of work	Professional Title	Organization	Years of experience
40	Male	Norway	Head of AI and user data	E24	23
37	Male	Norway	Frontansvarlig (Front Manager)	VG, Schibsted	17
42	Male	Sweden	Head of Editorial Development	Svenska Dagbladet	20
35	Female	Norway	Senior UX Researcher	Schibsted Media	6
40	Male	Norway	Head of AI and User Data	E24	23
45	Male	Sweden	Director of AI & Innovation	Aftonbladet	20+
39	Male	Sweden	Managing Editor	Omni	1
34	Male	Sweden	UX Lead	Aftonbladet	12
35	Male	Norway	AI Lead	Schibsted	15

Before interviews, participants received a written consent form clarifying study objectives, procedures, and voluntary participation in accordance with GDPR [11]. The form detailed data handling, anonymization, and deletion schedules, ensuring participants understood their rights. The interview protocol addressed four key themes: background and role, current practices and challenges in AI transparency, feasibility of adaptive transparency, and future recommendations. Full interview questions are provided in Appendix A (A.1–A.3), In parallel, the study reviewed internal user research, including a large-scale media trust survey of 3,000 participants (aged 16–74) in Sweden and Norway, conducted with NoA Consulting and Norstat. The survey operationalized user perspectives on trust, using brand-level questioning to reduce abstract responses and minimize social desirability bias. Respondents rated trustworthiness and media attributes on a 1–7 scale, and correlation analyses identified four key trust drivers: credibility of process, credibility of content, personal relevance, and selectivity [33]. These findings guided editorial transformation, transparency measures, and trust-driven media products, particularly in AI contexts.

Expert interviews were recorded, transcribed, and analyzed using a hybrid deductive-inductive coding process [16, 26]. Deductive coding, guided by established literature on AI transparency, editorial oversight, and adaptive UX [1, 2, 6], ensured the analysis was anchored in theoretical frameworks. This was followed by inductive coding to capture novel insights emerging from the expert narratives. All coding was conducted by the primary researcher, with oversight and discussion with colleagues to validate interpretations. Responses were grouped by expert category, and codes were aggregated into overarching themes forming the thematic analysis, which informed design requirements reflecting variations in media literacy, trust in AI, and expectations regarding AI-generated journalism [9, 16].

Throughout, a reflexive journal documented emerging assumptions, biases, and interpretations, enhancing analytic transparency. From this exploration, a five-pillar conceptual framework on the "paradox of disclosure" was developed, distinct from the three design frameworks guiding the prototype. It articulates how transparency can unintentionally erode user trust, capturing tensions identified in the literature and expert data [10, 15, 26, 27]. The five pillars are:

1. **Human Expectations** – audiences often equate journalism with human authorship, and AI disclosure disrupts this baseline credibility [7].
2. **Verifiable Sourcing** – providing audit-ready references mitigates distrust, but vague explanations fail; experiments show that source citations alongside AI labels can help recover trust [10].
3. **Cognitive Fatigue** – overloading readers with technical details can overwhelm and reduce trust, as highlighted by transparency paradox literature [27].
4. **AI Literacy** – low public understanding of AI makes dense explanations confusing; research shows that familiar language and basic AI literacy cues are essential [3].
5. **Editorial Control** – disclosures that obscure human oversight risk shifting accountability; transparency must clarify not only that AI was used but also how editorial responsibility was applied [2].

These pillars guided the design of adaptive transparency solutions, which adjust disclosure depth to user needs rather than assuming "more transparency always equals more trust" [8].

3.2 Phase 2: Ideation, Concept Exploration, and Prioritization

Following the foundational research, a structured ideation and wireframing process was employed. Prior to ideation, insights from expert interviews and a literature review were consolidated on a Miro whiteboard to create a foundation for the creative phase [22, 25]. During ideation, divergent thinking techniques such as speculative scenarios [22], brainstorming [28], and mind mapping [29] were used to generate a wide range of ideas. This transitioned into wireframing, where low-fidelity sketches were created and iteratively refined into high-fidelity prototypes in Figma, informed by the "User Experience 3.0" framework [9]. To ensure continuous improvement, iterative cycles of internal review and heuristic evaluation were adopted. Heuristic evaluation has been shown to identify usability issues early and cost-effectively [13, 14]. Progressive disclosure techniques were integrated to balance transparency with cognitive load [20].

A crucial component was the systematic prioritization of features using a value–effort matrix [37]. Each proposed feature was evaluated against the five pillars of the paradox of disclosure to produce a value score. Final feature prioritization employed an impact-minus-effort model, ensuring that selections for prototype development were both strategically aligned and resource-aware. This structured approach, drawing from design thinking [30], usability engineering [31], and adaptive UX frameworks [9, 21] provided a defensible method for translating research insights into actionable design interventions. Based on the prioritized concepts, three distinct interactive prototypes were developed to explore specific facets of adaptive UX

strategies for AI transparency. The development process was guided by principles of human-centered AI [2] and adaptive UX, which tailors interfaces to user context [8]. Creating multiple prototypes targeting different loci of transparency control enabled a richer understanding of where adaptive transparency can be most effective (see Table 2) [1].

Table 2: Prototype summary

Prototype identifier	Purpose	Key adaptive transparency feature	Link to paradox pillars
P1:CMS Disclosure Builder	Empower editorial control over AI disclosures	AI impact percentage, editable text, source links, "substantial impact box," humanization cues	Editorial Control; Verifiable Sourcing; Human Expectations
P2: Chatbot ("Hej Afton")	Explore transparency within conversational AI	"Show Thinking," personalization settings (tone, topics, sources), feedback interface	AI Literacy, Verifiable Sourcing, Editorial Control
P3: Content Settings Panel	Give users control over consumption format	Format switch (text/audio/video), in-article focus topics	Cognitive Fatigue

3.2.1 Prototype 1: CMS Disclosure Builder.

Figure 1 shows the AI playground, which houses a disclosure builder Integrated into a newsroom CMS it is designed to be integrated into the existing Omni CMS This prototype guides editors in crafting AI disclosures (see Figure 2). Key features include an AI Impact Percentage to quantify AI influence [6], Editable Disclosure Text for context-specific communication [8], optional Source Links to bolster verifiability [19], a "Substantial Impact Box" to emphasise editorial oversight [1], and Humanization Cues (e.g., bylines) to anchor trust in human authorship [2]., Figure 3 presents four embedded states of the tool as implemented in the AI Playground. In the first state, the toggle control activates the Disclosure Builder and displays the quantified AI Impact. The second state shows the available disclosure options that editors can select or deselect, including the model used, human review description, sources, and the "Substantial Impact" box. The third state highlights the "Substantial Impact" box itself, which contextualizes the degree of AI contribution and can be expanded through a dropdown. Finally, the fourth state provides a CMS preview of the composed disclosure output, enabling editors to review how transparency information will appear to readers before publication. Taken together, these embedded views demonstrate how the Disclosure Builder supports progressive, editor-controlled AI disclosures while preserving oversight (see Figure 3). appear to readers

before publication. Taken together, these embedded views demonstrate how the Disclosure Builder supports progressive, editor-controlled AI disclosures while preserving oversight (see Figure 3).

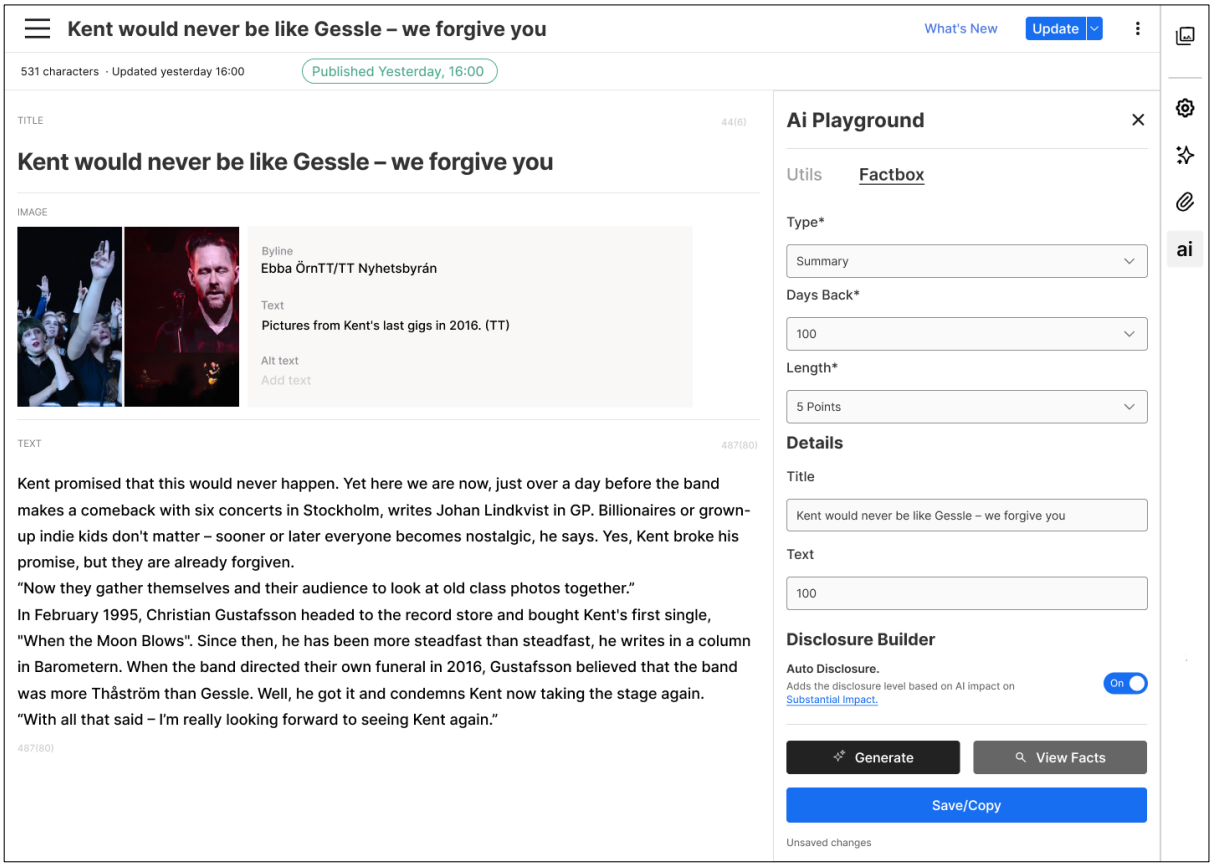


Figure 2: Disclosure Builder integrated into the existing Omni's Content Management System

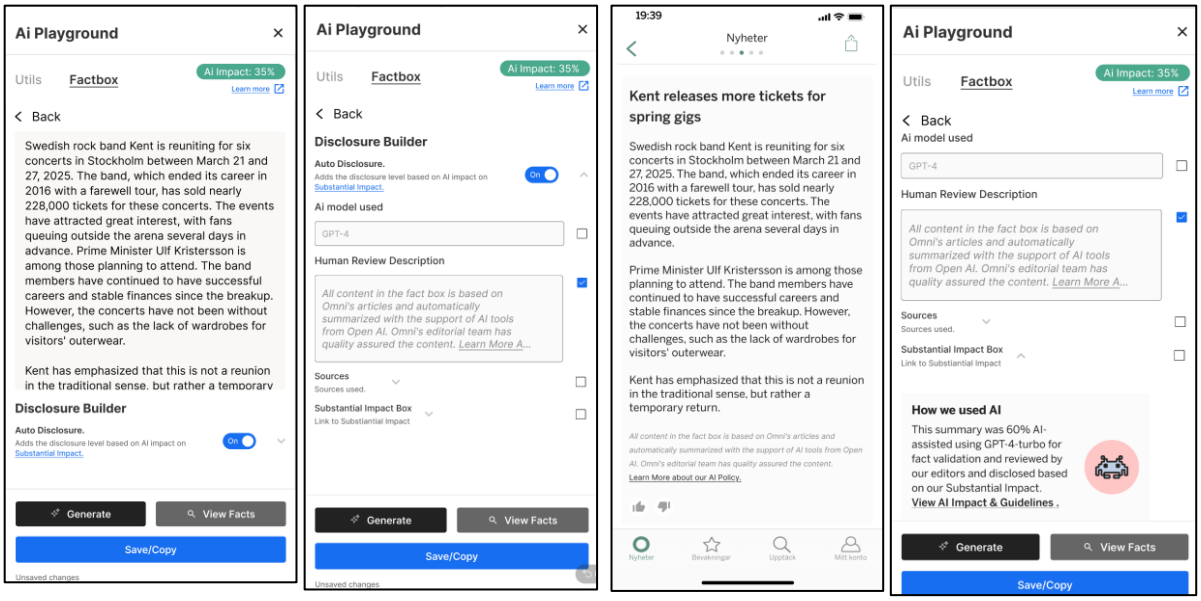


Figure 3: How the disclosure builder works and the output for users in the end.

3.2.2 *Prototype 2: Chatbot ("Hej Afton")*

A conversational interface with optional transparency controls. Figure 4 shows a “Show Thinking” button to reveal AI reasoning, aligning with explainable AI research [6]. Figure 5 shows “Personalization Settings” for tone, topics, and source preferences consistent with adaptive hypermedia principles [8] and source filters and feedback mechanisms to underpin trust and editorial control [2].

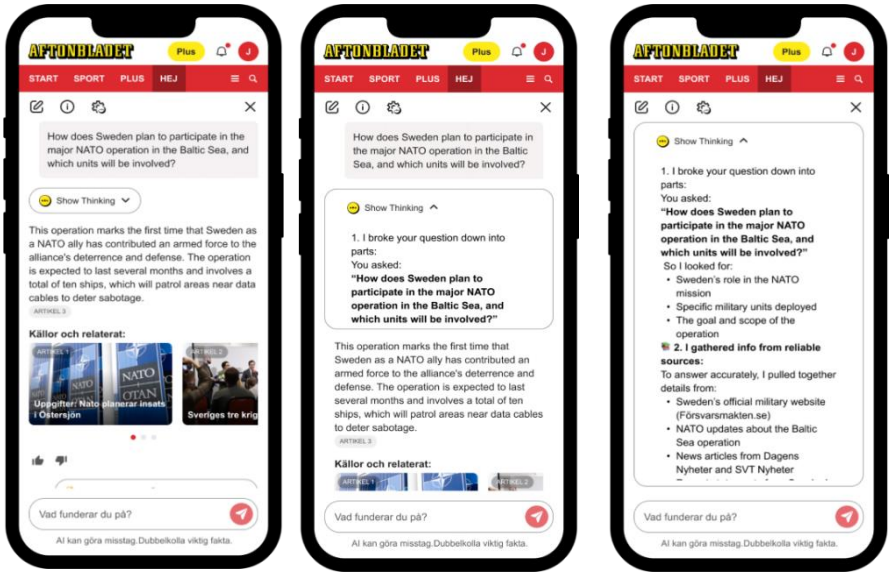


Figure 4: A “Show Thinking” button to reveal AI reasoning

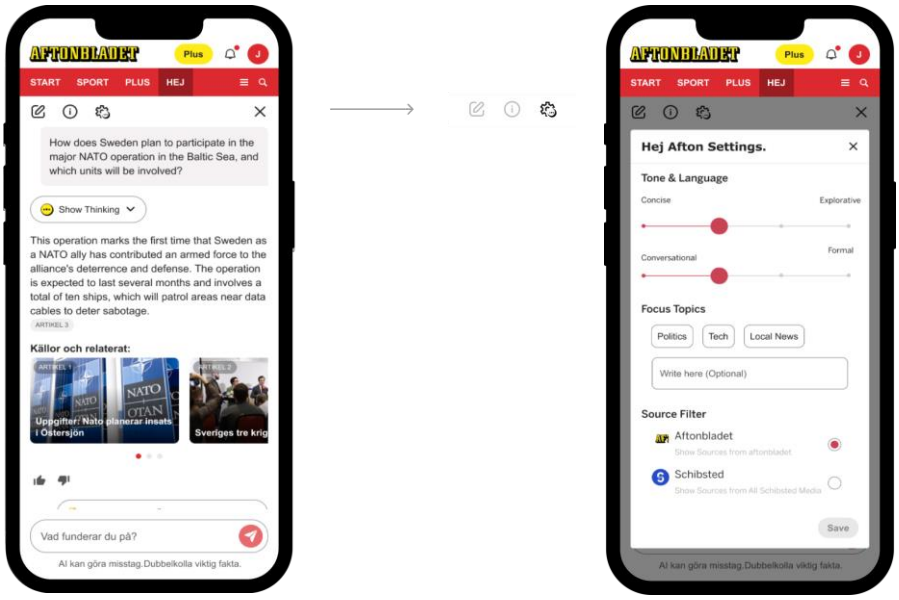


Figure 5: Personalization Settings for tone, topics, and source preferences are consistent.

3.2.3 *Prototype 3: Content Settings (Preference Panel)*

Allow users to tailor how news content is consumed. Figure 6 shows key features which include format flexibility (text/audio/video) to support accessibility and counter cognitive overload [9] and In-article Focus Topics to allow targeted user control and reduce information fatigue [1]. These prototypes collectively embody an adaptive transparency approach, aiming to mitigate the paradox of disclosure while preserving trust and usability [21].

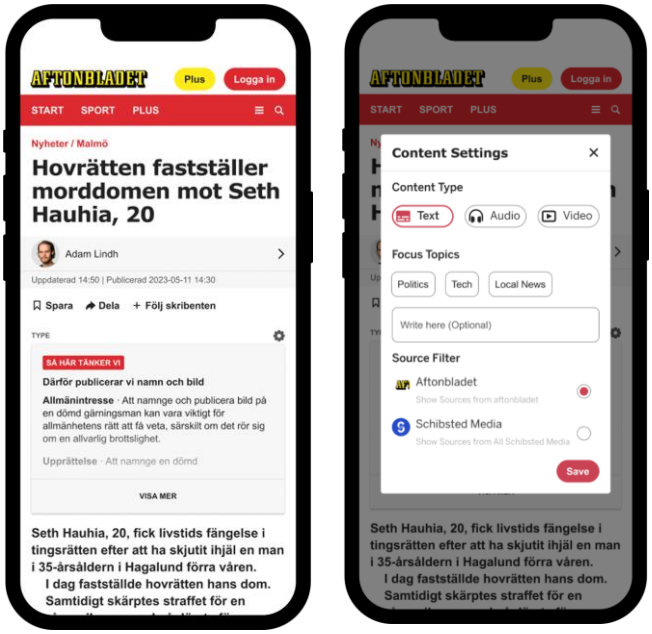


Figure 6: Content Settings

3.3 Phase 3: Evaluation of Adaptive Transparency Prototypes

To assess the three adaptive transparency prototypes (CMS Disclosure Builder P1, Chatbot with "Show Thinking" P2, and Personalized Content Settings Panel P3), three complementary evaluations were conducted. First, a heuristic inspection based on Nielsen's principles [13, 14] was carried out by a UX expert to examine usability, discoverability of controls, and the visibility of provenance cues. This helped identify issues such as overly technical labels and hidden settings that reduced clarity. Second, a think-aloud protocol was conducted with both the UX expert and a professional editor to capture real-time reflections while using the prototypes [5]. These sessions provided insights into how disclosure text, reasoning trails, and personalization options were interpreted in practice, and how they fit into editorial workflows. Third, a short-term value alignment checklist was applied to rate each prototype against the five pillars of the paradox of disclosure [10]. Ratings were given on a structured 0–2 scale, which highlighted the relative strengths of the CMS Disclosure Builder and the chatbot in contrast to weaknesses in the Content Settings Panel. Together, these methods validated design assumptions and revealed concrete usability and transparency challenges within editorial workflows.

4 RESULTS

This chapter presents a comprehensive analysis of the three prototypes through four complementary methods: Heuristic Evaluation, Think-Aloud Protocol, and the Short-Term Value Alignment Checklist. Together, these methods reveal each design's usability and transparency strengths and weaknesses, and their alignment with the Paradox of Disclosure [10].

4.1 Heuristic Evaluation

The Heuristic Evaluation, conducted by a UX expert using Nielsen's heuristics [13], assessed usability and transparency issues across all prototypes. For the CMS Disclosure Builder, several issues were identified. The label "AI Impact: 68%" was considered too technical and abstract, leading to a recommendation for simpler alternatives such as "Mostly AI-written" that would better align with editors' and readers' mental models [4]. Key settings were often hidden, forcing users to recall rather than recognize where controls were located, which reduced discoverability. Although a toggle for progressive disclosure was included, its outcomes were ambiguous, making it unclear to users what level of detail would be revealed. Provenance information was present but visually inconspicuous, requiring stronger placement. Despite these issues, editorial override functions worked well and ensured human-in-the-loop control, which was evaluated positively. For the Chatbot prototype, the heuristic review highlighted minor challenges in AI literacy. Some technical terms and personalization controls required explanation, and the lack of tooltips or plain language risked confusing users unfamiliar with AI concepts. Nevertheless, the toggle-based "Show Thinking" interaction was praised for allowing optional transparency without overwhelming the user, balancing detail with usability. The Content Settings Panel did not fare well in the heuristic evaluation. Its design required

users to carry a complete "mental model" of past and future use cases, which imposed a significant cognitive burden. Features such as "Focus Topics" were conceptually vague, leaving readers uncertain about whether content was being reframed or filtered. The absence of embedded source links undermined verifiability, while the abstract consolidation of settings created further complexity.

4.2 Think-Aloud Protocol

The Think-Aloud Protocol, conducted with both the UX expert and a professional editor, generated contextualized feedback on how the prototypes were experienced in practice [5]. For the CMS Disclosure Builder, the default disclosure text was clear and appropriately detailed, while the ability to trace sources supported verifiability, though the display of numerous numbered references was considered cluttered. The ambiguous thumbs-up/down mechanism highlighted the need for more precise feedback prompts, while adding a humanizing journalist byline next to AI disclosures emerged as a powerful trust signal [7, 9]. The "Substantial Impact Box" was useful for high-impact stories but redundant in low-impact contexts. Showing the reviewing journalist's name was recommended to reinforce accountability. For the Chatbot prototype, the "Show Thinking" feature was consistently highlighted as a strong step toward transparency. Personalization settings, including tone and topic filters, were engaging but required explanations [26]. Source filters could conflict with journalistic policy, and the thumbs-up/down feedback mechanism was ambiguous. The Content Settings Panel received mixed reactions. Format-switching options (text, audio, video) were intuitive, but the "Focus Topics" selector created confusion. Advanced settings should be deferred to reduce cognitive load [5]. The absence of editor-facing controls weakened oversight, and source filtering raised concerns about conflicting editorial standards. Overall, the abstract nature of the panel was cognitively heavy, and missing disclaimers or source links highlighted significant transparency gaps.

4.3 Short-Term Value Alignment Checklist

The Short-Term Value Alignment Checklist was used to systematically assess how each prototype addressed the five pillars of the Paradox of Disclosure [10]. Ratings were given on a scale from 0 to 2, where 0 = not at all addressed, 1 = partially addressed, and 2 = fully addressed. The results are summarized in Table 3. The CMS Disclosure Builder performed the strongest overall, fully addressing Human Expectations, Editorial Control, and Cognitive Fatigue (2/2 each), while partially addressing AI Literacy (1/2) and fully addressing Verifiable Sourcing (2/2). The Chatbot prototype ("Hej Afton") achieved full scores for AI Literacy and Verifiable Sourcing (2/2 each), partially addressed Human Expectations (1/2), and was rated as presenting no issue for Cognitive Fatigue (0/2). By contrast, the Content Settings Panel was weakest, with Human Expectations and Verifiable Sourcing not addressed at all (0/2), and AI Literacy and Cognitive Fatigue only partially addressed (1/2). Across these evaluations, five recurring themes emerged: technical terminology impeded comprehension (Human Expectations), hidden controls elevated Cognitive Fatigue, progressive disclosure was effective only in the CMS Disclosure Builder, sourcing was strongest in the Chatbot but weakest in the Content Settings Panel, and user-driven source filters risked creating tensions between editorial judgment and reader preferences.

Table 3: Results of the Short-Term Value Alignment Checklist across prototypes

Prototype	Human Expectations	Editorial Control	Cognitive Fatigue	AI Literacy	Verifiable Sourcing
CMS Disclosure Builder	Fully (2/2)	Fully (2/2)	Fully (2/2)	Partially (1/2)	Fully (2/2)
Chatbot ("Hej Afton")	Partially (1/2)	Fully (2/2)	No Issue (0/2)	Fully (2/2)	Fully (2/2)
Content Settings Panel	Not at all (0/2)	Partially (1/2)	Partially (1/2)	Partially (1/2)	Not at all (0/2)

(0 = not at all addressed, 1 = partially addressed, 2 = fully addressed).

5 DISCUSSION

5.1 Summary of results

Taken together, the evaluations highlight a central paradox: transparency features can both build and undermine trust, depending on how they are implemented [13]. The CMS Disclosure Builder consistently emerged as the most balanced design, combining editorial control, progressive disclosure, and clear mechanisms for introducing AI transparency. This includes the recommendation to quantify AI's "Substantial Impact" as a structured approach for evaluating the extent of AI involvement in news stories. Unlike ad hoc editorial impressions, this metric provides a consistent and auditable framework for assessing how much of an article has been shaped by AI. Percentage-based labels such as "AI Impact: 68%" were found too technical, while fully qualitative descriptions lacked clarity; the metric can therefore be operationalized in reader-friendly categories such as "Mostly AI-written" or "Edited by humans" [7]. By providing both editors and readers with a scalable and transparent standard, Substantial Impact mitigates bias, strengthens accountability, and anchors newsroom trust practices in measurable indicators [3]. The Chatbot prototype contributed important innovations, particularly the "Show Thinking" feature, enhancing explainability by presenting a high-level reasoning trail and improving AI literacy more effectively than static labels [4]. Editorial safeguards are needed around personalization controls to avoid inconsistencies. The Personalized Content Settings Panel, by contrast, revealed more risks than benefits: its ambiguous settings and lack of source verifiability increased cognitive load without delivering proportional transparency gains [13].

The evaluation underscores the Paradox of Disclosure: both excessive and insufficient transparency can undermine user trust. Embedding progressive layers of AI disclosures within familiar editorial workflows, as demonstrated by the CMS Disclosure Builder, enables context-sensitive transparency without overwhelming readers. A concise "AI-assisted" cue by default, with deeper provenance details available on demand, aligns disclosure formats with newsroom routines and audience feedback [3]. However, even well-designed disclaimers may induce fatigue if users lack clear controls for adjusting the level of detail. Without progressive gating—such as initial summaries that expand into full reasoning logs—detailed explanations risk confusing or discouraging non-expert users [13, 26]. The Personalized Content Settings Panel further highlights the risks of abstract configuration controls. While format-switching options (text, audio, video) were intuitive, buried and ambiguously labelled topic and source filters created uncertainty about AI's influence and increased cognitive load [13]. This demonstrates that autonomy must be balanced with editorial accountability, and transparency controls should surface contextually at the moment readers question content reliability, rather than residing in detached settings screens. From these insights, four design strategies emerge for effective adaptive transparency:

- **Objective impact benchmarks:** Disclosures should rest on quantifiable measures co-created with editorial and user research teams.
- **Progressive disclosure:** Begin with minimal visible cues and layer deeper explanations and source links only upon user request [14].
- **Humanized editorial cues**: Bylines or avatars alongside AI disclosures reinforce human oversight [3].
- **Contextual guidance**: Just-in-time tooltips, micro copy, or inline info icons help users navigate transparency layers without interrupting the reading experience [13].

Meaningful AI transparency thus arises from a human-centred, iterative integration of explainability into newsroom tools. As regulatory frameworks such as the EU AI Act evolve [11], organizations adopting adaptive, user-driven transparency strategies anchored in clear benchmarks, layered disclosures, humanized cues, and contextual support will be best positioned to maintain reader trust and uphold editorial integrity in an AI-augmented future.

5.2 Future work

Building on our findings, future research should prioritize extensive user testing with a diverse cross-section of everyday readers to capture not only cognitive usability metrics but also emotional and perceptual responses to AI disclosure features. Leveraging established usability methods such as think-aloud protocols [25] and Likert-scale surveys [33], alongside affective computing techniques (e.g., facial expression analysis), will reveal how transparency cues such as expandable "How this was made" panels impact trust, fatigue, and engagement across demographic groups. Combining A/B experiments [24] and comparing different disclosure formats [24] with in-depth qualitative feedback will enable newsrooms to tailor explanations that resonate emotionally while avoiding overload. Second, it is essential to embed and iteratively refine adaptive UX strategies within real newsroom workflows, rather than in isolated prototypes. This entails collaborating continuously with journalists,

editors, and product teams to integrate features such as progressive disclosure and chain-of-thought logs directly into content management and publishing tools. Longitudinal studies that track key performance indicators (e.g., time to publish, correction rates, audience retention) will measure the ongoing impact of adaptive transparency on both editorial efficiency and reader satisfaction, ensuring that design evolves in step with operational needs [3]. Third, further research must operationalize and validate our proposed framework for quantifying the "substantial impact" of AI assistance. Drawing on objective metrics such as text-edit ratios, summarization accuracy, and named-entity recognition fidelity, and combining these with audience perception data, will help balance editorial judgment with reader expectations. Statistical validation techniques, including reliability analysis and inter-rater agreement, can surface and mitigate human biases in AI oversight, reinforcing accountability and auditability [19, 24]. Finally, future work should explore how to weave transparency settings seamlessly into the natural user journey, rather than isolating them within a single settings screen. Guided by progressive disclosure principles [14] and usability heuristics [13], designers can experiment with in-context toggles, unobtrusive "info" icons, and contextual tooltips that appear exactly when users encounter AI-generated or AI-curated content. By aligning transparency controls with user tasks such as highlighting a sentence and then offering a "Why this?" expander, newsrooms can deliver on-demand insight without disrupting reading flow or introducing excessive cognitive load.

5.3 Contributions

This thesis makes four primary contributions to the field of human–computer interaction and journalism studies. First, it advances our conceptual understanding of the Paradox of Disclosure [10]. by empirically demonstrating how different transparency strategies ranging from simple AI labels to detailed chain-of-thought explanations simultaneously increase trust and introduce complexity that can undermine usability. By mapping these trade-offs across three distinct prototypes, we provide a nuanced framework for designers and newsroom product teams to navigate the tension between transparency and cognitive load. Second, the work offers a methodological contribution through its multi-method evaluation framework, which integrates heuristic inspection, think-aloud protocols, and a short-term value-alignment checklist. This triangulated approach not only yielded rich qualitative insights but also produced a replicable evaluation protocol that future researchers can apply to other adaptive transparency contexts [14, 25, 26]. The combination of expert review and real-world editorial feedback ensures that both usability heuristics and journalistic imperatives inform design decisions. Third, on a design level, the three prototypes, CMS Disclosure Builder, Chatbot with "Show Thinking," and Personalized Content Settings Panel, embody concrete instantiations of adaptive transparency principles such as progressive disclosure [26], explainable AI reasoning trails [4], and humanized accountability cues [3]. These artifacts serve as practical blueprints for integrating AI transparency into existing newsroom workflows, offering guidelines for embedding policy links, quantifying AI impact, and humanizing disclaimers without disrupting content production. Finally, this thesis delivers actionable recommendations for news organizations. By quantifying AI's "substantial impact," strengthening editorial identity within CMS interfaces, and adopting on-demand transparency mechanisms, the study outlines a road map for balancing user empowerment with institutional integrity. These insights are directly translatable to newsroom product road-mapping and can inform ongoing regulatory compliance efforts, such as those prompted by the EU AI Act [11].

5.4 Limitations

Despite its contributions, this research has several limitations that warrant consideration. First, the evaluation relied on high-fidelity prototypes rather than fully deployed newsroom systems, which may limit ecological validity. Although think-aloud sessions and heuristic reviews provided valuable formative feedback, real-world deployment could surface integration challenges not captured in a lab setting. Second, the participant pool was relatively small and homogeneous, comprising a limited number of UX experts and professional editors. While this expert sample yielded in-depth insights into editorial workflows, it may not represent the perceptions and needs of general news consumers. Future studies should include a broader demographic cross-section to validate emotional and trust responses across varied user segments [12, 25]. Third, the quantitative metrics proposed for measuring AI's substantial impact such as text-edit ratios and summarization accuracy remain conceptual and were not empirically validated within this thesis. Operationalizing and statistically testing these metrics will be essential for establishing reliable benchmarks and for balancing editorial judgment with audience expectations [24, 33]. Finally, the research context anchored in collaboration with a Nordic media organization may limit generalizability to newsrooms with different technological infrastructures, editorial cultures, or regulatory environments. While the core design principles are broadly applicable, localized adaptations may be necessary to address specific organizational constraints and

audience literacies. These limitations highlight opportunities for future research to extend and strengthen the adaptive transparency framework, ensuring its relevance across diverse journalistic ecosystems.

5.5 Deliverables and Integration of Schibsted Insights

This thesis produced three key deliverables, each grounded in user-centred design practices and informed directly by insights from Schibsted's internal research, stakeholder interviews, and newsroom-specific transparency challenges. First, a comprehensive set of design requirements was developed through a combination of expert interviews, heuristic evaluations, and literature synthesis. These requirements reflect both the functional and ethical dimensions of transparency in journalism, including needs for explainability, contextual disclosure, editorial override mechanisms, and clarity of language. They draw heavily from Schibsted's internal survey data and product reflections on audience preferences for AI-related messaging [34, 35, 36]. Second, an interactive prototype was created to demonstrate the practical application of adaptive transparency principles in editorial contexts. This prototype comprising three modules (CMS Disclosure Builder, Chatbot with "Show Thinking," and Personalized Content Settings Panel) showcases the use of progressive disclosure, chain-of-thought explainability, and personalization-aware controls. The design was iteratively refined based on Schibsted's product feedback and usability testing, ensuring that the interface remains aligned with real editorial workflows and user expectations. Third, the project produced a set of actionable design guidelines for integrating adaptive transparency measures into live newsroom systems. These guidelines synthesize theoretical findings from literature and empirical insights from this study and offer practical implementation strategies such as how to label AI-generated content in CMS platforms, when to surface AI impact disclosures, and how to balance personalization features with editorial integrity. Schibsted's internal transparency experiments and audience segmentation data played a significant role in shaping these recommendations, ensuring their alignment with ongoing organizational priorities and Nordic market sensibilities [3]. Together, these deliverables serve both as concrete outputs and as strategic enablers for product innovation within AI-enhanced journalism workflows. They are designed to support both near-term prototyping and long-term product development across Schibsted's brands and editorial platforms.

6 CONCLUSION

This study demonstrates that AI transparency in journalism must transcend simplistic labels and embrace adaptive, explainable strategies to cultivate user trust. Our evaluation of three prototypes the CMS Disclosure Builder, the Chatbot with "Show Thinking," and the Personalized Content Settings Panel—illuminated the "Paradox of Disclosure," wherein both excessive and insufficient transparency can erode credibility [10]. We found that effective AI transparency hinges on quantifying AI's role in content production, employing progressive disclosure to manage cognitive load, providing clear explanations of AI processes, humanizing disclaimers through editorial identifiers, and ensuring auditability via accessible provenance links.

Based on these insights, six actionable recommendations for newsrooms were offered. First, enhance language clarity by replacing technical jargon with plain-language descriptions and embedding concise explanations of AI features, thereby lowering the barrier to comprehension for general audiences [13, 18]. Second, improve feature discoverability through interface redesigns that surface key settings and personalization options prominently avoiding buried menus so that users can locate transparency controls intuitively [13]. Third, strengthen source transparency by clearly indicating content provenance with accessible links to original articles, data sources, and AI-policy documents, reinforcing the verifiability of AI-assisted outputs [18]. Fourth, balance disclosure and usability by optimizing the amount and presentation of transparency details: default to minimal, high-level disclosures with optional "More Info" expanders to prevent cognitive overload while preserving on-demand depth [13, 26]. Fifth, align editorial and user controls by developing mechanisms that permit user personalization such as format switching or topic preferences without compromising editorial standards or introducing conflicts of interest [7, 9]. Finally, conduct iterative testing with diverse user groups to continuously refine transparency features and validate that interface improvements meet evolving audience needs and expectations. In sum, adaptive transparency in AI-augmented journalism is a dynamic design challenge that requires a human-centered, iterative approach. By integrating quantifiable metrics, layered disclosures, humanized accountability cues, and rigorous user testing, news organizations can uphold both trust and usability navigating the Paradox of Disclosure to support an informed and engaged readership.

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A APPENDICES

Three scripts were followed for the interviews, tailored to the expertise of the participants.

A.1 Interview Questions for AI Transparency Experts

1. Can you tell me a bit about your work?
2. How do you work with AI transparency and its disclosure, and what challenges do you face?
3. How do user-facing AI models that the organization is working with handle transparency, and what limitations exist in disclosing AI decision-making?
4. What do you think is the most critical information for audiences to know about AI-generated content?
5. How do you avoid "transparency fatigue" while ensuring users receive essential information?
6. Are there risks in making AI transparency too detailed or complex for general audiences to understand?
7. Some argue that AI transparency should be adapted based on audience characteristics. What are the technical and ethical trade-offs in doing this?
8. From a technical perspective, how feasible is it to create adaptive transparency mechanisms in AI-driven journalism?
9. Do you think personalized AI transparency should exist, and why?
10. What recommendations would you give to news organizations seeking to improve AI transparency today?

A.2 Interview Questions for Journalists & Editors.

1. Can you tell me a bit about your work and the kind of journalism you specialize in?
2. How frequently does your publication use AI-generated content, and if so, how are they using it and in what contexts?
3. From your perspective, how has AI changed journalistic workflows and reporting standards?
4. How do your readers or news consumers typically react to AI-generated content? Have they raised any concerns or trust issues?
5. What current methods does your newsroom use to disclose AI-generated content? How do these compare to traditional transparency methods for human-written content?
6. Are there any industry-wide best practices you follow when presenting AI-generated content?
7. Do you believe different audience segments (e.g., younger vs. older readers, tech-savvy vs. non-tech-savvy, etc) have varying expectations for AI transparency? If so, would you shed some light on them?
8. Have you observed differences in how younger audiences interact with AI-generated content compared to older readers?
9. What kind of information do you think is most important for audiences to know about AI-generated content (e.g., source attribution, level of human oversight, AI model details), and how should it be presented visually?
10. How does AI transparency affect their credibility as professionals now and how/if you think it will change in the future if we personalize AI transparency?

A.3 Interview Questions for UX Designers & Product Managers

1. Can you tell me about your background and the types of products you typically design or research?
2. Can you describe your role and experience with designing/researching interfaces for AI-driven content?
3. Have you worked on projects where AI transparency was a design consideration? If so, how was it approached, and what challenges did you encounter?
4. How does your platform currently communicate AI-generated content to users?

5. Have you conducted or reviewed user research on how audiences perceive AI-generated content? What key insights emerged?
6. What UX patterns have you found most effective for communicating AI transparency?
7. How do you balance providing enough transparent information without overwhelming users (avoiding "transparency fatigue")?
8. Do you think AI transparency should be personalized based on user characteristics, or should it be standardized for all users?
9. What factors (e.g., age, digital literacy, media trust levels) seem to influence how users interact with AI transparency features?
10. Have you seen successful examples of personalized transparency in other domains (e.g., finance, healthcare) that could be applied to journalism?

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